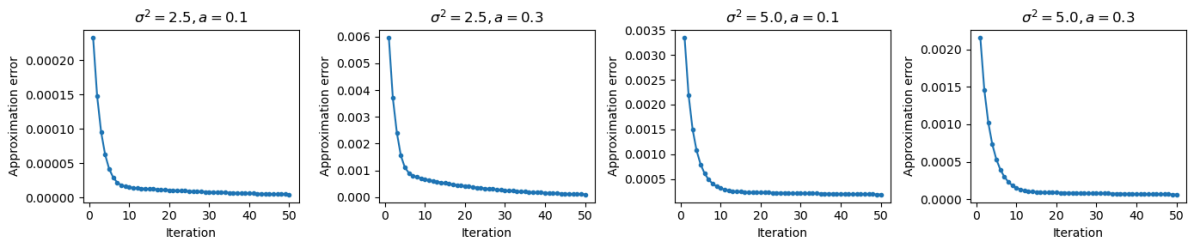


Result Reproduction

Since the original experiment did not provide publicly available code, the simulation study in this work was conducted based on our own reproduced implementation of the proposed algorithm.

To ensure that the implementation logic is consistent with the original study, we first reproduced the main experimental results reported in the original paper (Chapter 5). Using the same parameter settings, we evaluated the approximation error of the distributed EM estimator relative to the pooled estimator during the first 50 EM iterations under different signal-to-noise ratios σ^2 and heterogeneity levels a , with sample size $n = 1000$ and number of sites $K=10K = 10K=10$. The results are shown below:



The result is approximately the same as Figure 1 in Chapter 5.

I then reproduce the test in the simulation section. The same code logic and parameter settings were used. The evaluation indicators and visualization results are as follows.

